

An Offline Intelligent Vietnamese Input Method Based on Deep Learning

Capstone Project Report 1 · AIP491

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NextKey Overview

Redefining Vietnamese Mobile Typing

The Continuous Latin Input Paradigm

Traditional IMEs (Telex/VNI) require typing explicit tone marks and accents. NextKey introduces a continuous Latin input paradigm (similar to Chinese Pinyin).

Three Core Technical Pillars

- **Continuous Unaccented Input:**
Type text without spaces or tone marks (e.g., "toidihoc" → "toi di hoc").
- **End-to-End Deep Learning:**
Jointly predicts word boundaries and restores diacritics in a single real-time forward pass.
- **100% Edge-Based Execution:**
Fully offline execution with latency < 50 ms and absolute user data privacy.

Ergonomic Efficiency

Keystroke Savings (KS):

$$KS = \left(1 - \frac{|X_{\text{continuous}}|}{|X_{\text{Telex}}|} \right) \times 100\%$$

- **20%–40% reduction** in screen taps.
- Eliminates explicit spacebar & tone keys.
- Lowers thumb strain & RSI injury risks.

Project Motivation & Societal Value

Three Foundational Pillars of Impact

1. Language Inclusion

Language Scaffold for All:

- **Children & Students:**
Type thoughts freely without getting stuck on complex spelling rules.
- **Elderly Users:**
Eliminate cognitive strain of memorizing tone key combinations on small screens.
- **Foreign Learners:**
Express Vietnamese naturally without diacritical barriers.

2. Ergonomics & Speed

Maximize Efficiency:

- **20%–40% KS Savings:**
Significantly reduces physical typing effort.
- **Continuous Flow:**
Eliminates manual spacebar and accent key presses.
- **Health Impact:**
Reduces thumb fatigue and Repetitive Strain Injury (RSI) risks.

3. Seamless Multilingual

Zero "Dính Dâu" Errors:

- **No English Corruption:**
Prevents unwanted accents in words like "case" → "cae", "deadline", or "meeting".
- **Zero Layout Switching:**
Seamlessly type mixed Vietnamese-English sentences without toggling input modes.

Input Method Paradigms & Keystroke Comparison

Quantitative Example of Continuous vs. Traditional Typing

Typing Paradigm	User Input Keystrokes	Restored Output	Keystrokes
Standard Telex	t-o-o-i-space-d-d-i-space-h-o-c-j	toi di hoc	13 taps
Standard VNI	t-o-6-i-space-d-9-i-space-h-o-c-5	toi di hoc	13 taps
NextKey (Proposed)	t-o-i-d-i-h-o-c	toi di hoc	8 taps (38.5% KS)
NextKey (With Typos)	t-o-i-d-i-h-o-c-b-a-k-k	toi di hoc bai	12 taps ($k \rightarrow i$)

Core Advantages

- Continuous typing flow without manual space insertion.
- Seamless multilingual typing (no English tone pollution).
- Robust against adjacent-key fat-finger errors.

Deployment Constraints

- **Inference Latency:** < 50 ms per event (fluid 60fps).
- **Memory Footprint:** < 2 MB INT8 compressed model.
- **Privacy Guarantee:** 100% offline client-side execution.

Literature Review & Baseline Systems

Assessment of Existing Architectural Approaches

Model Family	Representative	Strengths	Critical Gaps / Limitations
Rule-Based IMEs	Unikey, EVKey	Deterministic, zero RAM, instant speed.	Rigid; zero typo tolerance; requires explicit mode toggling for English terms.
Statistical N-Grams	HMM, Kneser-Ney	Fast inference; effective for local context.	Exponential state space; fails on unsegmented continuous sequences.
Recurrent Networks	Bi-LSTM + CRF	Captures sequential character context.	Slow autoregressive decoding; historically decoupled diacritics from typos.
Pretrained LLMs	PhoBERT, ViT5, ByT5	SOTA semantic representation.	Large footprint (> 100M); high latency (> 300ms); infeasible for edge.

Mathematical Problem Formulation

Formal Definition of Multi-Task Sequence Transformation

Sequence Transformation Formulation

Given noisy unaccented $X = (x_1, \dots, x_T) \in \mathcal{V}_{\text{input}}^T$, find clean target $Y = (y_1, \dots, y_M) \in \mathcal{V}_{\text{output}}^M$:

$$\hat{Y} = \arg \max_Y P(Y | X; \theta) = \arg \max_Y \prod_{t=1}^M P(y_t | y_{<t}, X; \theta)$$

Compound Noise Taxonomy ($\mathcal{T}$)

- **Diacritic Stripping:** Removal of all 6 tones.
- **Whitespace Erasure:** Word merging.
- **QWERTY Spatial Typos:** Neighbor keys.
- **Residual IME Codes:** Incomplete Telex.

Multi-Task Objective

$$\mathcal{L}_{\text{total}} = \lambda_{\text{diac}} \mathcal{L}_{\text{diac}} + \lambda_{\text{seg}} \mathcal{L}_{\text{seg}} + \lambda_{\text{typo}} \mathcal{L}_{\text{typo}}$$

- Joint cross-entropy optimization for diacritics, boundaries, and typos.

Core Research Questions (RQ1 – RQ6)

Foundational Scientific Inquiries Driving NextKey

Modeling & Noise Inquiries

- **RQ1 (Restoration Feasibility):**
Can a compact network restore text from compound noise (accents, spaces, typos)?
- **RQ2 (Architecture Pareto Trade-off):**
How do non-autoregressive taggers compare against Seq2Seq decoders in speed vs. accuracy?
- **RQ3 (Noise Calibration):**
Does QWERTY spatial noise modeling improve real-world out-of-domain generalization?

Edge Optimization & UX Inquiries

- **RQ4 (Quantization & Distillation):**
Can KD + INT8 quantization achieve > 95% teacher parity with < 50ms latency?
- **RQ5 (Candidate Selection Utility):**
Does Top- k candidate generation improve practical typing Keystroke Savings (KS)?
- **RQ6 (Privacy-Preserving Edge):**
Can on-device adaptation learn user vocabulary with zero remote server logging?

Project Objectives (S.M.A.R.T.)

Target Research & Engineering Milestones

Data & Modeling Objectives

- **Corpus Curation:** Build multi-domain parallel dataset ($> 840K$ sentences, JDWR v1) across 8 distinct genres.
- **Noise Engine:** Implement 6-dimensional QWERTY spatial and colloquial noise generator.
- **Multi-Task Learning:** Train candidate models for joint diacritic, spacing, and typo restoration.

Edge Optimization & User Targets

- **Model Compression:** Distill and quantize model to INT8 (< 2 MB, < 50 ms on CPU).
- **Ergonomic Savings:** Validate 20%–40% Keystroke Savings (KS) over standard Telex.
- **Zero Leakage:** Guarantee 100% offline client execution with zero network telemetry.

Project Scope & Deliverables

Clear Operational Boundaries

In-Scope Deliverables

- End-to-end deep neural sequence restoration models.
- Keyboard-aware synthetic noise generator pipeline.
- Quantized ONNX/C++ edge inference API.
- Interactive proof-of-concept web application.

Out-of-Scope Boundaries

- Low-level OS kernel keyboard drivers (iOS/Android virtual keyboard system integration).
- Voice recognition & multimodal speech input.
- Historical scripts (Hán-Nôm) or minority dialects.

Team Structure & Research Tracks

Clearly Partitioned Roles and Responsibilities

Role & Track	Assignee	Core Research & Engineering Responsibilities
Project Lead & AI Researcher	Tran Dinh Gia (SE181550)	• Project coordination, sprint reviews, milestone tracking. • Sequence modeling mathematical formulation & loss design. • Architecture exploration and training supervision.
Data Engineering & Preprocessing	Truong Nhat Quang (DE180544)	• Multi-domain parallel corpus collection (> 840K) and cleaning. • Synthetic QWERTY spatial noise & abbreviation pipeline. • In-domain and Out-of-Domain evaluation dataset curation.
Edge Systems & Deployment Lead	Nguyen Xuan Cuong (DE180528)	• Knowledge Distillation (KD) & INT8 quantization engines. • Edge hardware profiling and latency benchmarking (< 50ms). • Offline inference backend API and interactive web interface.

Work Breakdown Structure (WBS) & Timeline

20-Week Five-Phase Execution Plan

Phase	Phase Name	Key Work Packages & Deliverables	Duration	Lead
Phase 1	Data & Literature Foundation	Parallel corpus collection (> 840K sentences), synthetic typo & spacing noise engine, tokenization vocabulary schema.	Weeks 1–4	Quang
Phase 2	Model Development & Baselines	Candidate architecture training (BiGRU, T5, Transformer taggers), multi-task loss tuning, validation error analysis.	Weeks 5–9	Gia
Phase 3	Optimization & Quantization	Knowledge Distillation (KD) to compact Student, INT8 Post-Training Quantization (PTQ), WER/BLEU & latency benchmarking.	Weeks 10–14	Cuong
Phase 4	Backend & Integration	Offline C++/ONNX inference engine, REST/WebSocket key-board API, stress testing, interactive demo prototype.	Weeks 15–17	Cuong
Phase 5	Validation, Thesis & Defense	Human user testing, Keystroke Savings (KS) validation, comprehensive capstone report, and defense presentation.	Weeks 18–20	All

Risk Management Framework

Risk Register and Proactive Mitigation Matrix

Identified Risk		Severity	Prob.	Proactive Mitigation Strategy
Domain	Generalization Gap	High	Med	Incorporate realistic synthetic noise (QWERTY typos, chat slang, abbreviations); benchmark against isolated OOD domain (<i>the_thao</i>).
High Inference Latency		High	Med	Apply Knowledge Distillation and INT8 quantization; prioritize non-autoregressive sequence taggers over heavy autoregressive decoders.
Compute Resource Limits		Med	High	Leverage distributed dual GPU cloud allocations (Kaggle Dual T4, Google Colab); utilize mixed-precision training (FP16).
Keystroke	Privacy Concerns	High	Low	Enforce 100% client-side offline execution; strictly zero remote data transmission or telemetry logging.

Quality Assurance, Evaluation & Decision Gates

Rigorous Multi-Tier Validation Methodology

Hierarchical 3-Tier Metric Harness

- **Character-Level:** CER, Typo Recovery, Boundary F1.
- **Word-Level:** WER, Word Accuracy, Token F1.
- **Sentence-Level:** Exact Match (EM), Near-Perfect ($\leq 5\%$), BLEU-4, ROUGE-L.
- **Ergonomic Metric:** Keystroke Savings (KS).

Scientific Decision Gates

- **Seed Protocol:** Fixed global seed 42 across all splits.
- **Gate 1 (Smoke Run Gate):** Reject architectures failing loss convergence within 1,000 steps.
- **Gate 2 (Objective Benchmark Gate):** Decisions strictly justified by quantitative metric gains on validation sets.
- **Budget Constraint:** \$0.00 capital expenditure.

Summary of Expected Deliverables & Key Indicators

Key Performance Targets

- **Inference Latency:** < 50 ms per sentence on CPU.
- **Keystroke Savings:** 20%–40% reduction vs. Telex.
- **Model Size:** < 2 MB footprint via INT8 quantization.
- **Restoration Fidelity:** SOTA BLEU-4 & Word Accuracy.
- **Security:** 100% offline client-side edge inference.

Tangible Capstone Deliverables

- Open-source reproducible codebase & dataset pipeline.
- Quantized model weights & ONNX/C++ runtime.
- Interactive web demo & comprehensive thesis report.

Thank You for Your Attention!

Questions & Discussion Welcome